

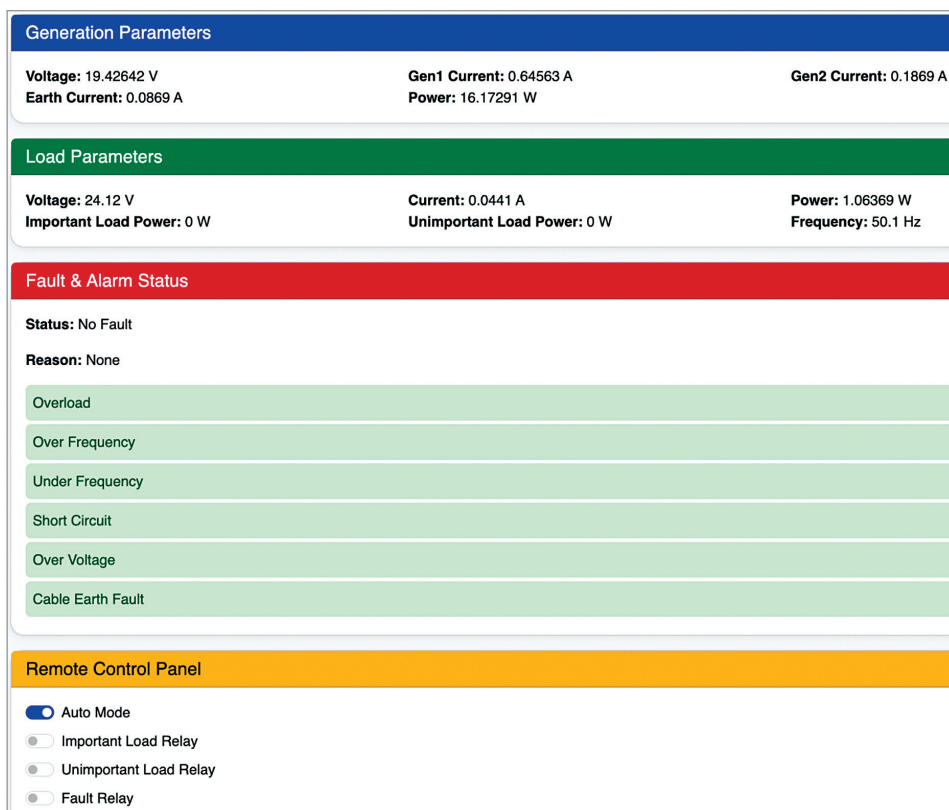


Fig. 8: Real-time monitoring dashboard

Step 3: Enable Firebase Services Realtime Database

1. Go to Build→Realtime Database
2. Click Create Database
3. Select location
4. Choose Start in Test Mode (for ESP32 testing)

Authentication (If using login/anonymous login)

1. Go to Build→Authentication
2. Click Get Started
3. Go to Sign-in Methods
4. Enable Anonymous Sign-in

For the generator-side ESP32 code, begin by setting the Wi-Fi SSID and password to the local network settings. Configure the API key in

the code, followed by the database URL. After configuration, select the appropriate ESP32 board version and upload the code. Fig. 5 shows the generator-side Firebase API configuration.

For the load-side code, follow

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the same sequence: configure the Wi-Fi credentials, enter the API key, and configure the database URL. Once the settings are complete, upload the code by selecting the NodeMCU ESP8266 board.

Fig. 6 shows the configuration of the load-side code snippet.

Construction and testing

After uploading the code, the ESP32 can connect to a Wi-Fi network via a mobile hotspot. Once the connection is established, the Serial Monitor will display a message similar to the one shown in Fig. 7.

After powering the AC supply, feed it into the 230V/18V transformer. Real-time data will then appear in the Firebase Realtime Database Console.

The console enables monitoring and manual control of the load, including switching off the fault relay when necessary. To view real-time data on the web application, visit <https://esp32camsecurity.web.app> **EFY**

Murugananda Bharathi S. (left) is Executive Engineer at the Tamil Nadu Electricity Board with a strong interest in IoT-based energy management and electrical electronics. Jenishaa Bharathi M. (right) is an electronics and IoT enthusiast pursuing B.Tech in Computer Science with specialisation in Artificial Intelligence at Amrita University.

Sherin Bharathi M. is an engineer with interests in coding, smart technologies, and advanced electronics. She earned her Integrated B.Tech – M.Tech in Data Science from IIT Madras.



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